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Isotopic Dating and Geomagnetic Polarity Studies on Volcanic Rocks from Mauritius, Indian Ocean

IAN McDOUGALL and F. H CHAMALAUN

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Isotopic Dating and Geomagnetic Polarity Studies on Volcanic Rocks from Mauritius, Indian Ocean

Abstract: Results of a geochronological and paleomagnetic investigation of the volcanic island of Mauritius are reported. Potassium-argon dates indicate that the main shield volcano was built subaerially between about 7.8 and 6.8 m.y. ago in the early Pliocene. Some evidence is presented for caldera formation following the construction of the volcanic shield. Profound erosion subsequently destroyed much of the volcano, leaving only peripheral steep-sided massifs. Lavas of the Younger Volcanic Series were erupted from about 3.5 m.y. ago until less than 0.2 m.y. ago, flooding the denuded stumps of the shield volcano. This activity was intermittent and a hiatus is recognized from about 2 to 0.7 m.y. ago, which we have taken as the break between the eruption of the Early and Late Lavas of the Younger Volcanic Series.

The mean directions of magnetization of the three groups of lavas are not significantly different from one another and are close to an axial dipole direction. The combined paleomagnetic and dating studies on the lavas also yield further information on the geomagnetic polarity time scale; 16 results on rocks with ages lying between 3.5 and 0.17 m.y. are consistent with the established scale for this period of geological time. Of particular significance are results that confirm the Mammoth reversed event at 3.0 m.y. ago, and the age of the boundary between the Gauss normal and Gilbert reversed polarity epochs at 3.35 m.y. Nineteen results were obtained on rocks whose ages are between 4.6 and 7.9 m.y., and these data generally are consistent with the polarity time scale based upon extrapolations from the sea-floor spreading hypothesis.

The evolution of the Mascarene Islands, comprising Mauritius, Réunion and Rodriguez is briefly reviewed and it is concluded that each island had developed independently, and that no correlation of age with respect to distance from the Mid-Indian Ocean Ridge is observed. The volcanism that has built Mauritius and Réunion may be a continuation of the development of the Mascarene Plateau with migration of the activity generally southward with time.

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INTRODUCTION

In terms of the continental drift hypothesis the Indian Ocean resulted from fragmentation and movement of the southern continents. Hence studies of the evolution and age of oceanic islands in this region are of some importance in understanding the development of the Indian Ocean. For this reason, geochronological and paleomagnetic investigations of the volcanic rocks of Mauritius, Réunion and Rodriguez, comprising the Mascarene Islands and situated in the western Indian Ocean, were commenced by us in 1964. Some results for Réunion and Rodriguez have already been published (Chamalaun and McDougall, 1966; McDougall and others, 1965). In this paper we report data obtained on rocks from Mauritius, and briefly review the geological evolution of the Mascarene Islands.

These studies were also made to provide further information on the geomagnetic polarity time scale, which, from combined K-Ar dating and paleomagnetic investigations on volcanic rocks, is now known in considerable detail for the last 3.6 m.y. (McDougall and Chamalaun, 1966; Cox and Dalrymple, 1967a). The pattern of reversals found is substantially confirmed by polarity determinations on deep-sea sedimentary cores (Opdyke and others, 1966; Watkins and Goodell, 1967). Using the sea-floor spreading hypothesis of Vine and Matthews (1963) to interpret the linear magnetic anomaly patterns adjacent to the ocean ridges, together with the polarity time scale as determined for the last 3.6 m.y., Heirtzler and others (1968) have extrapolated the polarity time scale back to about 80 m.y. ago. It is obviously desirable to extend the geomagnetic polarity time scale by direct K-Ar dating and polarity measurements on surface rocks to test these extrapolations. However, because the errors in K-Ar dating are more or less a fixed percentage of the age, it becomes more difficult to define the time scale adequately with increasing age. It is particularly difficult beyond about 6 m.y. ago, as pointed out by Cox and Dalrymple (1967a). Nevertheless, detailed studies on sequences of rocks older than 6 m.y. may provide sufficient information to test some of the predictions of Heirtzler and others (1968), as illustrated by the work of Baksi and others (1967), and this has also been attempted in Mauritius.

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We particularly thank J. de B. Baissac of Vacoas for assistance in many ways while one of us (McDougall) was in Mauritius. The late J. Vinson, formerly Director of the Mauritius Institute, kindly provided access to the facilities of the Institute. Some core samples were provided by R. Sentenac. This work was undertaken through the interest of Professor J. C. Jaeger, and we express our gratitude for his continual support. Technical assistance in the dating was ably given by I. H. Ingram and Miss S. Dinter, and H. Berry and E. Pederson assisted in rock processing. The K-Ar data were reduced and calculated by computer using a program developed by Mrs. T. Chamalaun.

GEOLOGY OF MAURITIUS

General Statement

The island of Mauritius is situated in the western Indian Ocean at long. 57° 30′ E. and lat. 20° 20′ S. It is approximately elliptical in outline with major axis (NNE.-SSW.) of 60 km, and minor axis about 45 km. Except for raised reefs and a fringing coral reef Mauritius is entirely volcanic.

Mauritius lies near the southern end of the Mascarene Plateau, a major, aseismic, shallow structure that rises abruptly from oceanic depths of more than 3000 m on either side, and extends in a gentle arc from the Seychelles Archipelago in the north over 2300 km toward the south (Fig. 1; Heezen and Tharp, 1964, 1965, 1966; Fisher and others, 1967). Detailed bathymetric investigations (Fisher and others, 1967) show that Mauritius is the summit of a somewhat larger mass that rises steeply from the ocean floor on all sides, except to the north, where the increase in depth is more gradual. The western flank of the island is steep and straight and may be fault controlled; the eastern

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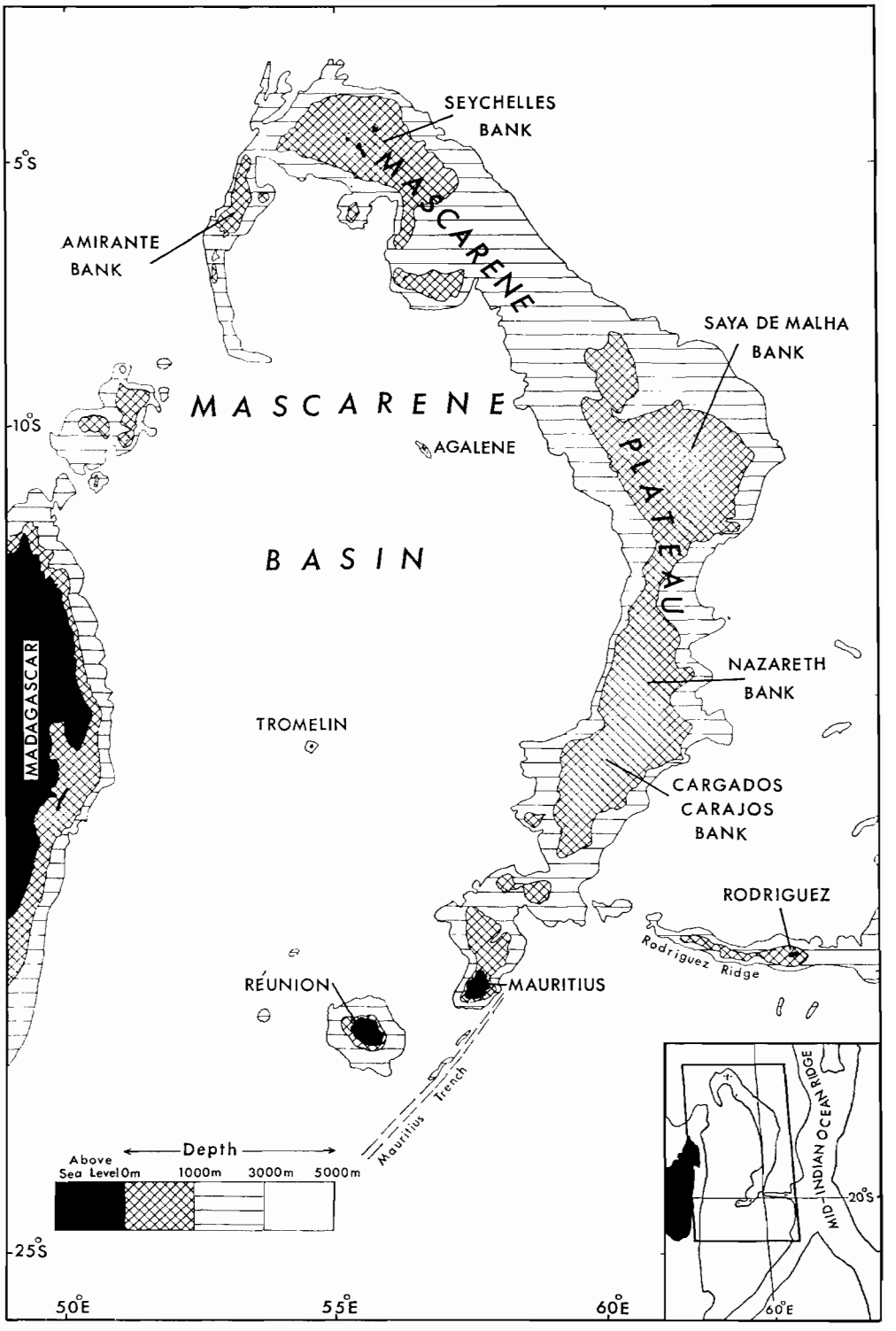


Figure 1. Generalized bathymetric map of the western Indian Ocean (*after* Fisher and others, 1967).

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flank is less regular and less steep. The steep flanks grade into a more gently sloping apron at depths greater than 2500 m, merging into the ocean floor at about 4000 m depth. To the southeast the apron is truncated by the straight, narrow and deep Mauritius Trench that has been traced for more than 1400 km in a southwest direction. Mauritius and its underlying mass appears to be a single volcanic edifice built up from oceanic depths.

The geology of Mauritius was described by de Chazal and Baissac (1949), Simpson (1951), and Walker and Nicolaysen (1954), and these authors summarize the earlier work. Simpson (1951) recognized an Older Volcanic Series and a Younger Volcanic Series; between the eruption of these two series there was a long period of erosion. The Younger Volcanic Series was subdivided into the Early and Late Lavas. The areal distribution of the rock series is shown in Figure 2, which closely follows the geological map of Simpson (1951).

Older Volcanic Series

Rocks of this series form the large massifs that occur discontinuously in an arc from near Port Louis in the northwest, along the west coast, and adjacent to the central east coast of Mauritius (Fig. 2). The massifs are built predominantly of basaltic lavas and agglomerates, generally dipping seaward at low angles. Locally, and particularly in the Port Louis massif, dips up to 15° are observed. As suggested by some of the earlier workers, the massifs, which commonly have steep inland flanks, probably represent the remnants of a large shield volcano that has been very deeply eroded, particularly in the central region. The lava flows are massive and range from 5 to 30 m in thickness. A great variety of rock types is present (Walker and Nicolaysen, 1954), although basaltic compositions predominate. The basalts range from aphyric to strongly phyric types, and include olivine basalt, picrite, ankaramite and feldsparphyric basalt. Alkali andesites also occur, and these and more fractionated lavas are particularly well developed in the Blanche Mountains where the lavas are essentially horizontal and may form a caldera filling. Chemically the lavas are undersaturated alkaline types, although they rarely show nepheline in the norm (Walker and Nicolaysen, 1954). Dikes of olivine basalt are common. At least five plugs or domes of trachyte and phonolitic trachyte, first correctly identified by Shand (1933), have been found in the Older Volcanic Series.

Younger Volcanic Series

Lavas of the Younger Volcanic Series flooded the much eroded stumps of the old volcano to form an extensive intramontane plain, which slopes away from a central ridge. Two major periods of eruption are recognized, thought to have been separated by a relatively short period of quiescence. The Early Lavas are now only exposed in southwest Mauritius (Fig. 2), but may well extend under the Late Lavas (Simpson, 1951). These Early Lavas are massive to somewhat vesicular alkali olivine basalts, usually 3 to 10 m thick. Petrographically and chemically they form a rather uniform series and are nepheline normative (Walker and Nicolaysen, 1954).

The Late Lavas cover the remainder of the island between the older massifs and were erupted from a chain of vents extending in a line striking north-northeast through Curepipe in the center of the island. The surface of the Late Lavas has been little modified by erosion, although a number of gorges have been cut by consequent streams. As recognized by all workers, the Late Lavas must be very young geologically. The lavas generally are thin (1 to 8 m) and highly vesicular, suggesting a rather volatile magma. They are entirely basaltic and contain abundant olivine phenocrysts. Many of the lavas are relatively coarse grained, doleritic basalts with a subophitic or ophitic texture. Compositionally the Late Lavas show little variation and are nepheline normative; they are characterized by remarkably low potassium contents for alkali olivine basalts (Walker and Nicolaysen, 1954).

GEOCHRONOLOGY

Methods

Samples were collected from lavas of both the Older and Younger Volcanic Series; localities are shown in Figure 2 and listed in the Appendix, together with petrographic notes where appropriate. Dating was undertaken by the K-Ar method mainly on whole rock samples that showed little or no alteration in thin section. The dating techniques used were described by McDougall (1964, 1966). Potassium was measured by flame photometry (Cooper, 1963), and the isotope dilution method was used for determining argon. A Reynolds-type mass spectrometer in the static mode of operation was employed to measure the isotopic ratios of the argon. Because of memory effects, extrapolation of the isotope ratios to zero time is necessary, and as the changes are

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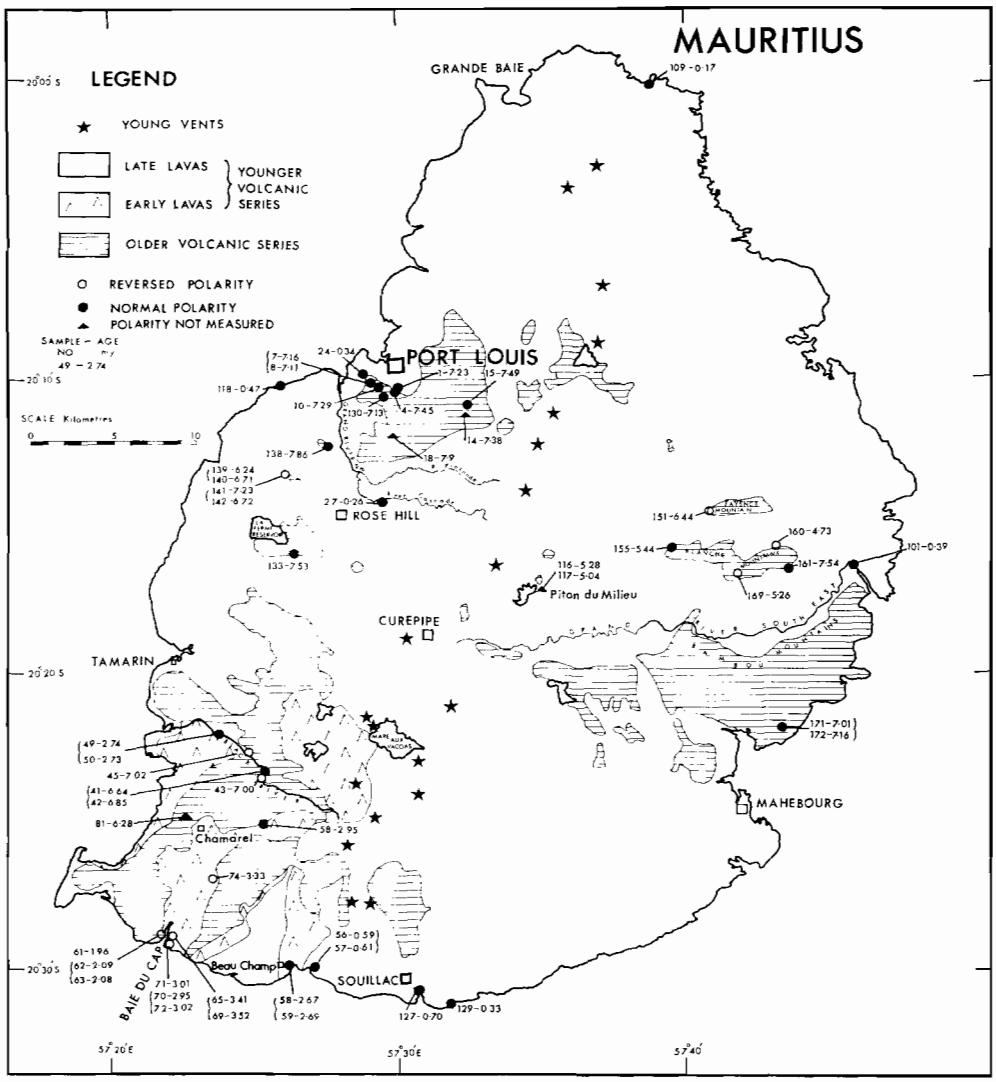


Figure 2. Geology of Mauritius (*after* Simpson, 1951) showing sample localities, age results and magnetic polarity.

linear in most cases, a least squares method was used. Orifice and discrimination corrections were applied to the zero time values of the ratios. Constants used for the calculation of age were:

$$\lambda_e = 0.584 \times 10^{-10} yr^{-1};$$
$$\lambda_\beta = 4.72 \times 10^{-10} yr^{-1};$$
$$^{40}K = 1.19 \times 10^{-2} \text{ atom percent.}$$

Estimates of the precision of the age measurements can be made in two ways. The first method is based upon analysis of variance where replicate determinations are available. Using the results from 14 samples in Tables 1 through 4 for which argon was measured more than once, and where the air correction was less than 82 percent, the pooled coefficient of variation is 1.37 percent for a single measurement. Combined with a coefficient of variation of 0.35 for potassium, an over-all precision for a single age is estimated to be 1.42 percent at the 68 percent level of confidence. The error is accountable almost entirely in terms of the uncertainty in the tracer calibrations. The

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second way of calculating the precision is by the statistical appraisal of each analysis, and the errors, at the 68 percent level of confidence, are quoted in Tables 1 to 4 using this method. The two methods give comparable estimates of the precision. The second method is similar in principle to those given by Baksi and others (1967) and Cox and Dalrymple (1967a), and allows for the error magnification effect as the proportion of air argon in the argon determinations increases.

In Table 6, where the age is calculated for multiple determinations from one or more samples, the mean age and its error are computed by assigning weights inversely proportional to the square of the errors of the individual age determinations (Topping, 1962).

Older Volcanic Series

Samples of the Older Volcanic Series were collected from most of the massifs, but because many lavas showed considerable alteration, relatively few samples were dated.

The lavas of the massif immediately to the east and northeast of Port Louis dip approximately 15° to the northwest, and were sampled in some detail along the track to Signal Mountain to obtain a stratigraphic sequence. These lavas were quite fresh and the dates obtained are listed in Table 1 in stratigraphic order. The results show excellent internal consistency and agree essentially to within experimental error. A mean age of 7.23 m.y. is indicated. The sample from Quoin Bluff is stratigraphically below the summit of Signal Mountain because of the dip of the lavas. The basalt from Anse Courtois is considerably lower in the sequence than the Signal Mountain lavas and its indicated age is in agreement with this. However, because of the high air contamination and the poor quality of these particular argon analyses, the apparent age difference may not be significant. The two samples from near Pieter Both, in the same massif, give similar ages to those found in the Signal Mountain section. These results indicate that the lavas of the Port Louis area were erupted over a relatively short period of time.

Samples from scattered localities in western and southwestern Mauritius (Table 2) give indicated ages similar to those found in the Port Louis massif. However, several exceptions need to be noted. Two samples (MU 139, MU 140) from the same alkali andesite lava flow from the Grand Malabar Mountains yield dates that differ by 7 percent, which is outside normal experimental error. Sample MU 139 has a slightly altered appearance in thin section, but apart from alteration of olivine to iddingsite, MU 140 is fresh; some loss of radiogenic argon from MU 139 may have occurred. The overlying basalt contains abundant large crystals of olivine, clinopyroxene and plagioclase, some of which occur in aggregates and are possibly xenolithic. Whole rock analyses on two samples (MU 141, MU 142) give K-Ar dates that also disagree by more than the expected errors, but the dates are similar to that found for MU 140. A much older date of 10.1 ± 1.6 m.y. was found for large plagioclase crystals separated from MU 142, reinforcing the view that these crystals may be xenolithic. Alternatively, the plagioclase may carry excess radiogenic argon (compare with Damon and others, 1967). The poorly reproducible results on MU 141 and MU 142 may be explained, at least in part, by this effect. Nevertheless, an age of about 7 m.y. for these lavas appears likely.

Another exception relates to the Chamarel Trachyte which intrudes the Older Series in southwest Mauritius. Anorthoclase phenocrysts separated from two samples collected about 300 m from one another gave quite different apparent dates. Sample MU 81 yielded an age of 6.3 m.y., which is consistent with the stratigraphy and other dates on the Older Volcanic Series, but MU 80 gave a highly aberrant age of 17.7 m.y. (Table 2). This appears to be a clear case of excess radiogenic argon in alkali feldspar at a level of about 6.5×10^{-11} moles/g, and it is inferred that these phenocrysts crystallized under a significant, but variable, pressure of radiogenic argon.

Lavas of the Older Volcanic Series in eastern Mauritius generally are altered; hence only a few samples were dated (Table 2). The dates found for basalts from Lion Mountain (7.1 m.y.) and the Blanche Mountains (7.5 m.y.) are similar to those obtained from lavas in the western part of the island. Above the basalt in the Blanche Mountains, and also to the north in the Fayence Mountains, occur massive, essentially horizontal lavas in which individual flows commonly exceed 15 m in thickness. These lavas are more strongly fractionated types (alkali andesites and trachyandesites, Walker and Nicolaysen, 1954), and possibly were ponded in some form of caldera (de Chazal and Baissac, 1949). The young ages found support this suggestion, but the large differences in apparent age between the indi-

TABLE 1. POTASSIUM-ARGON DATES ON WHOLE ROCK SAMPLES FROM THE PORT LOUIS MASSIF, OLDER VOLCANIC SERIES, MAURITIUS

GA No.	Field No.	Sample	Potassium Analyses (wt. percent)	Radiogenic Argon ($\times 10^{-11}$ moles/g)	Atmospheric argon-40 (percent)	Calculated Age (m.y.)	Locality
2144	MU 8	Alkali andesite	1.356,1.366	1.698 1.750	20.8 26.0	7.00 ± 0.10 7.22 ± 0.12	Top of Signal Mountain, altitude 305 m
2326	MU 7	Alkali andesite	1.389,1.390	1.771	31.8	7.16 ± 0.12	
2138	MU 130	Ankaramite	1.139,1.142	1.448	18.3	7.13 ± 0.07	Quoin Bluff, altitude 320 m
2115	MU 10	Olivine basalt	1.293,1.295	1.682	14.5	7.29 ± 0.15	Signal Mountain, altitude 220 m
2113	MU 4	Olivine basalt	1.051,1.065	1.404	57.7	7.45 ± 0.08	Signal Mountain, altitude 115 m
2325	MU 1	Olivine basalt	1.043,1.043	1.355 1.334	12.2 9.3	7.29 ± 0.13 7.18 ± 0.12	Signal Mountain, altitude 75 m
2116	MU 18	Basalt	1.099,1.106	1.534 1.588	88.7 92.6	7.81 ± 1.10 8.08 ± 1.80	Anse Courtois
3118	MU 14	Basalt	1.094,1.097	1.439	38.3	7.38 ± 0.14	Below Pieter Both
3119	MU 15	Basalt	0.944,0.946	1.260	40.4	7.49 ± 0.13	

TABLE 2. POTASSIUM-ARGON DATES ON SAMPLES FROM THE OLDER VOLCANIC SERIES IN WESTERN AND EASTERN MAURITIUS

GA No.	Field No.	Sample	Potassium analyses (wt. percent)	Radiogenic argon ($\times 10^{-11}$ moles/g)	Atmospheric argon-40 (percent)	Calculated Age (m.y.)	Locality
Western Mauritius							
2140	MU 138	Olivine basalt	0.932,0.933	1.306	70.0	7.86 ± 0.18	Petite Malabar
2143	MU 141	Basalt	0.549,0.552	0.709	76.1	7.23 ± 0.15	Same lava; Western end Grand Malabar Mountains
3137	MU 142	Basalt	0.485,0.486	0.581	75.6	6.72 ± 0.15	Same lava; Western end Grand Malabar Mountains
3137	MU 142	Plagioclase	0.070,0.071	0.127	88.2	10.08 ± 1.56	Same lava; Western end Grand Malabar Mountains
2141	MU 139	Alkali andesite	1.438,1.464	1.635 1,610	28.7 34.7	6.32 ± 0.11 6.16 ± 0.11	Same lava; Western end Grand Malabar Mountains
3136	MU 140	Alkali andesite	1.436,1.439	1.719	41.3	6.71 ± 0.11	Same lava; Western end Grand Malabar Mountains
2139	MU 133	Oceanite	0.878,0.880	1.179	74.6	7.53 ± 0.14	La Ferme Reservoir
2188	MU 41	Olivine basalt	0.903,0.907	1.069	42.0	6.63 ± 0.06	Same dike; Black River
2327	MU 42	Olivine basalt	0.897,0.897	1.071 1.112	35.9 36.8	6.71 ± 0.12 6.99 ± 0.12	Same dike; Black River
2119	MU 43	Olivine basalt	0.847,0.855	1.058 1.068	61.9 68.2	6.98 ± 0.07 7.04 ± 0.14	dike; Black River
2120	MU 45	Basalt	1.010,1.014	1.266	77.7	7.02 ± 0.02	Black River

TABLE 2. (*Continued*)

GA No.	Field No.	Sample	Potassium analyses (wt. percent)	Radiogenic argon ($\times 10^{-11}$ moles/g)	Atmospheric argon-40 (percent)	Calculated Age (m.y.)	Locality
2128	MU 81	Anorthoclase	4.52 ,4.49	5.041	24.9	6.28 ± 0.07	Chamarel Trachyte
3125	MU 80	Anorthoclase	4.51 ,4.51	10.32	16.5	17.75 ± 0.31	
Eastern Mauritius							
2133	MU 116	Anorthoclase	4.41 ,4.42	4.152	50.5	5.28 ± 0.05	Piton du Milieu Trachyte
2134	MU 117	Anorthoclase	4.40 ,4.41	3.956	34.2	5.04 ± 0.05	
2142	MU 160	Alkali andesite	1.918,1.926	1.622	26.4	4.73 ± 0.06	Hermitage, Blanche Mountains
2334	MU 169	Alkali andesite	2.260,2.262	2.119	22.4	5.26 ± 0.09	Mt. Barbe, Blanche Mountains
2332	MU 155	Trachyandesite	3.291,3.300	3.191	58.9	5.44 ± 0.14	West flank, Mt. Blanche
2333	MU 161	Basalt	0.903,0 918	1.222 1.223	30.6 23.9	7.54 ± 0.13 7.54 ± 0.13	Deep River, Blanche Mountains
2331	MU 151	Alkali andesite	1.394,1.395	1.606 1.594	77.7 79.6	6.47 ± 0.20 6.42 ± 0.13	Fayence Mountains
2335	MU 171	Olivine basalt	0.795,0.806	0.999	45.5	7.01 ± 0.12	Lion Mountain
2336	MU 172	Olivine basalt	0.814,0.815	1.039	26.3	7.16 ± 0.13	

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vidual flows is unexpected. This may be because of variable argon loss, as the rocks are fine grained and show some deuteric alteration (*see* Appendix). In contrast to the Chamarel Trachyte, anorthoclase separated from two samples of the trachyte at Piton du Milieu gave concordant dates of about 5.15 m.y., which is therefore believed to approximate the age of emplacement of this body.

Younger Volcanic Series

Early Lavas. These lavas, now only exposed in the southwestern part of Mauritius, probably were erupted from sources in the high country near Mare aux Vacoas (Simpson, 1951).

Good exposures of the Early Lavas occur at Baie du Cap, particularly on the east side of the bay where they dip gently to the south. The lowest exposed basalt here gave concordant ages of 3.45 m.y. on two samples (MU 65, MU 69, Table 3). Three samples from two lavas several flows higher in the succession yielded ages agreeing closely at 3.0 m.y. Although a difference in age of about 0.4 m.y. was not anticipated, the excellent internal consistency of the results suggests that the difference is real. Lavas on the west side of Baie du Cap gave consistently lower ages of about 2 m.y. (Table 3); these basalts were regarded in the field as probably equivalent to those exposed on the east side of the bay. Reconsideration of the field relationships now suggests that a narrow valley cut into the Older Volcanic Series was filled with lavas between 3.4 to 3.0 m.y. ago and that subsequently a new valley, marginal to the old one, was eroded and filled with 2.0 m.y. old lavas. Further erosion produced the present Baie du Cap valley.

That the Early Lavas cropping out north from Baie du Cap were erupted from a source to the north during the same period of time is supported by the dates of 2.95 m.y. on a lava from Plaine Champagne (MU 85) and 3.3 m.y. on a sample (MU 74) 4 km north of Baie du Cap. Slightly younger dates of about 2.7 m.y. were recorded for Early Lavas exposed at the junction of the Petite Gorges and Black River, adjacent to the west coast of Mauritius, and at Beau Champ on the south coast (MU 58, MU 59). Although the lavas at Beau Champ were mapped by Simpson (1951) as Late Lavas it is clear from the age and also the petrographic characteristics that they should be included in the Early Lavas; Figure 2 has been modified accordingly.

Late Lavas. The doleritic basalts of the Late Lavas of the Younger Volcanic Series are difficult to date because of their unusually low potassium contents and relatively high air contamination, which is partly related to the vesicular nature of these rocks. The few dates obtained range from 0.7 to 0.17 m.y. (Table 4). No internal checks can be made with so few results, but in view of the possibility that all pre-existing radiogenic argon was not completely outgassed during eruption and cooling, the younger ages in particular may be considered as maxima.

The earliest of the Late Lavas (mouth of Rivière des Galets and Souillac, on the south coast) are more massive basalts, distinct from the typical doleritic basalts of this group. Petrographically, they are more akin to the Early Lavas, but as they were mapped by Simpson as Late Lavas and have considerably younger ages than most of the Early Lavas, they are included within the younger formation. On this basis there is a hiatus of about 1.3 m.y. between the time of eruption of the Early and Late Lavas, consistent with Simpson's view that there was an erosional interval between the two formations.

Evolution of Mauritius

The potassium-argon dates confirm the general outline of the geological evolution of Mauritius as suggested by Simpson (1951) and earlier workers, and provide a time scale for the history of volcanism. In particular, the dates on lavas of the Older Volcanic Series in the steep-sided massifs provide convincing support for the idea that these massifs are the much eroded remnants of a single, large, shield volcano. The results show that the shield volcano was built above sea level by the extrusion of lavas over a relatively short period of time between about 7.8 and 6.8 m.y. ago in the early Pliocene, using the time scale of Funnell (1964). Subsequent to the main dome building phase, a caldera probably developed in the eastern part of the volcano, into which more fractionated lavas were erupted and ponded about 5.5 m.y. ago. The collapse may have been more widespread, particularly in the central parts of the volcano, but evidence for this is lacking owing to the widespread cover of the Younger Volcanic Series. The dates from the trachyte at Piton du Milieu suggest that this and other trachyte plugs and domes may have been emplaced during the same period of collapse and caldera formation.

TABLE 3. POTASSIUM-ARGON DATES ON WHOLE ROCK SAMPLES OF OLIVINE BASALTS FROM EARLY LAVAS OF YOUNGER VOLCANIC SERIES, SOUTHWEST MAURITIUS

GA No.	Field No.	Potassium Analyses (wt. percent)	Radiogenic Argon ($\times 10^{-12}$ moles/g)	Atmospheric argon-40 (percent)	Calculated Age (m.y.)	Locality
2124	MU 58	0.519,0.520	2.380	59.2	2.58 ± 0.03	Beau Champ
			2.528	54.3	2.73 ± 0.05	
			2.497	50.7	2.70 ± 0.05	
3122	MU 59	0.400,0.400	1.917	63.1	2.69 ± 0.05	
2121	MU 49	0.542,0.543	2.644	65.5	2.74 ± 0.03	Near junction of Petites Gorges and Black River, west coast
2122	MU 50	0.586,0.588	2.853	53.3	2.73 ± 0.03	
2328	MU 61	0.535,0.538	1.854	61.0	1.94 ± 0.04	West side of Baie du Cap
			1.885	41.3	1.98 ± 0.04	
3134	MU 62	0.506,0.506	1.882	38.9	2.09 ± 0.04	Same lava
2329	MU 63	0.530,0.531	2.010	51.2	2.13 ± 0.04	
			1.928	49.1	2.04 ± 0.04	
2127	MU 71	0.522,0.525	2.793	54.1	3.00 ± 0.03	East side of Baie du Cap
			2.812	53.2	3.03 ± 0.05	
3135	MU 70	0.494,0.497	2.601	67.9	2.95 ± 0.05	Same lava
3123	MU 72	0.538,0.538	2.891	54.7	3.02 ± 0.05	
2125	MU 65	1.181,1.212	7.266	50.4	3.41 ± 0.03	Same lava
2126	MU 69	1.124,1.125	7.045	65.2	3.52 ± 0.05	
3124	MU 74	0.719,0.720	4.241	43.1	3.31 ± 0.06	Baie du Cap Valley
			4.298	28.2	3.36 ± 0.06	
2129	MU 85	0.731,0.733	3.844	58.4	2.95 ± 0.03	Plaine Champagne

TABLE 4. POTASSIUM-ARGON DATES ON WHOLE ROCK SAMPLES FROM LATE LAVAS OF YOUNGER VOLCANIC SERIES

GA No.	Field No.	Sample	Potassium Analyses (wt. percent)	Radiogenic Argon ($\times 10^{-13}$ moles/g)	Atmospheric argon-40 (percent)	Calculated Age (m.y.)	Locality
2132	MU 109	Doleritic basalt	0.347,0.348	1.082 1.052	93.6 94.0	0.176 ± 0.015 0.171 ± 0.012	St. François, north coast
3120	MU 27	Doleritic basalt	0.285,0.287	1.317	92.0	0.26 ± 0.02	11 km south of Port Louis
2117	MU 24	Doleritic basalt	0.368,0.368	2.198	91.3	0.34 ± 0.01	3 km south of Port Louis
2135	MU 118	Doleritic basalt	0.365,0.366	3.056	95.4	0.47 ± 0.08	Pointe aux Sables, west coast
2131	MU 101	Doleritic basalt	0.216,0.217	1.767 1.360	97.0 97.9	0.46 ± 0.07 0.35 ± 0.05	Grand River South East
2137	MU 129	Olivine basalt	0.446,0.447	2.532 2.598	80.5 81.8	0.32 ± 0.02 0.33 ± 0.01	Gris Gris, south coast
2123	MU 57	Olivine basalt	0.532,0.535	5.79 5.91	60.8 68.1	0.61 ± 0.01 0.62 ± 0.01	Mouth, Rivière des Galets
3121	MU 56	Olivine basalt	0.526,0.526	5.54	69.1	0.59 ± 0.02	
2136	MU 127	Olivine basalt	0.483,0.488	5.96 6.08	89.8 85.9	0.69 ± 0.03 0.70 ± 0.02	Souillac

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Subsequently, profound erosion of the shield volcano occurred, destroying much of the central region and leaving only the peripheral steep-sided massifs, themselves strongly dissected. About 3.5 m.y. ago, in the late Pliocene, there was renewed volcanic activity which produced the Younger Volcanic Series. These eruptions continued intermittently until very recent times. The volcanism was of quite different character to that displayed during the building of the shield volcano in that eruption probably occurred from relatively small vents. Although the lavas of the Younger Volcanic Series are very widespread, the volume is relatively small in comparison with the earlier activity. As recognized by Simpson (1951) the younger lavas were erupted in two main phases. The Early Lavas were extruded intermittently from about 3.5 to 2.0 m.y. ago and are exposed only in the southwestern part of Mauritius. They may have been much more extensive; possibly they underlie the very widespread Late Lavas which flooded the topography between about 0.7 and 0.17 m.y. ago and cover about three-quarters of the island. Taking the age of the Pliocene-Pleistocene boundary as about 2 m.y. (McDougall and Stipp, 1968), the Early Lavas are late Pliocene, whereas the Late Lavas are Pleistocene. Simpson (1951) suggested that the Late Lavas probably ceased eruption more than 100,000 years ago because these basalts are locally overlain by raised coral reefs assigned to high Pleistocene sea levels; our dating results confirm this estimate.

Simpson (1951), although fully recognizing the difficulties in estimating age, tentatively proposed that the building of Mauritius by volcanism from the ocean floor began in the Cretaceous, and continued into the early Tertiary. Our results indicate that the massifs, the eroded remnants of the main shield volcano, took about a million years to build. As the present maximum altitude of the massifs is about 1000 m, by extrapolation the whole edifice, which rises some 5000 m above the ocean floor, could have been built in 5 million years. These approximate calculations indicate that the construction of the volcano from the ocean floor may have commenced only about 12 m. y. ago in the late Tertiary, rather than in the early Tertiary or Cretaceous.

PALEOMAGNETISM

Samples were collected from all three rock groups and studied paleomagnetically. The samples were orientated with a Brunton compass and four cylinders were cut from each sample. Initially, the N.R.M. directions of two specimens from each sample were measured and on the basis of these results 20 representative samples were selected for alternating field demagnetization up to 600 oersted (peak field). The a.f. demagnetization results for the test specimens showed that a peak field of 225 oersted would be sufficient to remove unstable components of magnetization, and two or more specimens from each sample were subsequently cleaned in this field. These cleaned results are used in this paper.

In Figure 3 the a.f. curve of the magnetically hardest and softest sample from each rock group are shown. The terms magnetically "hard" and "soft" are used here to describe the amount of N.R.M. intensity associated with low coercivities. A magnetically soft sample loses its intensity of magnetization relatively rapidly during a.f. demagnetization and is therefore potentially more unstable than a magnetically hard rock. Figure 3 shows that the Early Lavas contain the hardest samples and the Older Volcanic Series the softest, but there is considerable overlap between the two groups and both cover a wide range. The Late Lavas are intermediate in character and lie within a rather narrow range. The a.f. characteristics of the Mauritius volcanics are similar to those found for Réunion lavas (Chamalaun, 1968).

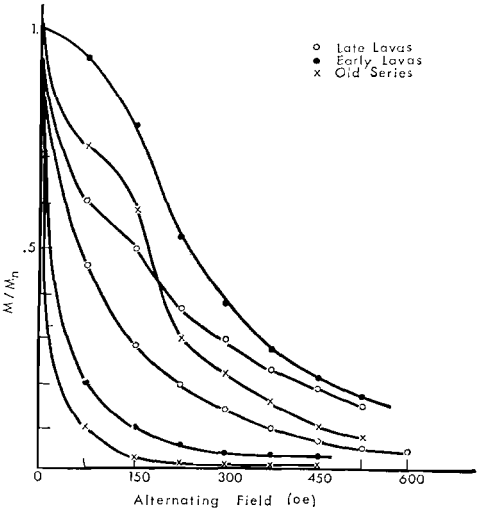


Figure 3. Alternating field demagnetization curves for Mauritius rocks.

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Some thermal demagnetization curves are shown in Figure 4. Only a few specimens were thermally demagnetized, so it is not certain whether the difference between the Late Lavas and Older Series apparent in the figure is real. However, it is noteworthy that low Curie points (~ 300°C) are predominant as was found for the Réunion lavas (Chamalaun, 1968).

As was the case in Réunion, the detailed isotopic dating results show that the samples are unevenly distributed in time and fall into the three distinct age groups discussed above. No difficulty was encountered in allocating the few undated samples to these groups. In Figure 5 the flow mean directions (giving unit weight to sample means) are plotted for the three rock groups.

The Late Lavas of the Younger Volcanic Series are all normally magnetized and the between flow scatter is relatively small. The mean inclination, −45° (Table 5), is higher than that expected from an axial dipole field (−37°), whose direction lies outside the 95 percent error circle of the observations. The discrepancy is probably due to inadequate sampling of the secular variation and is therefore regarded as not significant.

Four flows of the Early Lavas are normally magnetized and the remainder are reversed. One of the normal flows has an abnormal direction (277.1° E., −78.4°), but as the mean is based on five internally consistent samples from the same flow, it must be assumed to be a true direction. The flow was dated at 2.75 m.y., which places it very close to the Kaena reversed event (McDougall and Chamalaun, 1966). It is therefore possible that the direction is dominated by the non-dipole field at the time of a polarity reversal. The two reversed flows in Figure 5c with high inclinations occur on the west side of Baie du Cap and were dated at 2.0 m.y. (MU 61, MU 62, MU 68). The flows on the east side have low inclinations as does the flow at the head of the valley, all of which were dated at 3.0 to 3.5 m.y. old. Both normal and reversed polarities are also present in the Older Volcanic Series. The samples span about 3 m.y. with the majority lying between ages of 6.5 and 7.5 m.y. The scatter in the directions is rather large and does not reduce even if higher a.f. fields are used.

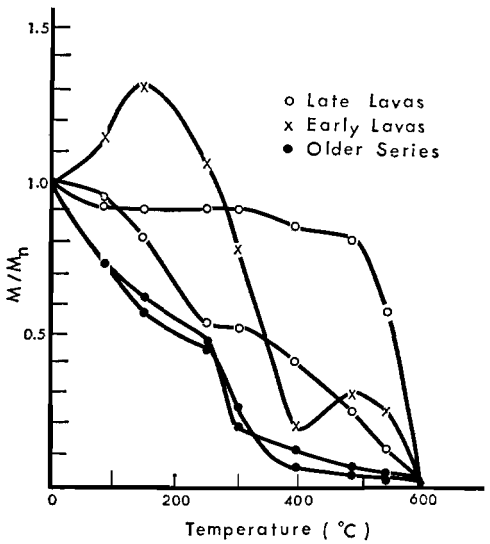


Figure 4. Thermal demagnetization curves for Mauritius rocks.

In Table 5 the results for the three groups are subdivided according to the polarity and means are formally computed for each. Not too much significance should be attached to these means as the secular variation is not adequately represented due to the uneven time distribution of the samples. Nevertheless, the mean directions are not significantly different from one another or from an axial dipole field.

GEOMAGNETIC POLARITY TIME SCALE

The main reason for combining paleomagnetic and isotopic dating studies on the volcanic rocks of Mauritius was to obtain more information on the frequency and duration of the polarity changes of the geomagnetic field. The Mauritius and Réunion results are summarized in Figure 6 and Figure 7. Table 6 summarizes the relevant magnetic and isotopic data, on a flow average basis for Mauritius. The 15 results obtained from Mauritius with ages less than 3.5 m.y. are entirely consistent with the geomagnetic polarity time scale discussed by McDougall and Chamalaun (1966) and Cox and Dalrymple (1967a).

The seven flows with normal polarity and ages between 0.25 and 0.70 m.y. further reduce the possibility that an event of reversed polarity within this age range has remained undetected. The results from sample MU 127 also support the age suggested for the Bruhnes-Matuyama epoch boundary. No young flows were encountered that might have confirmed the event at 0.01 m.y. as tentatively suggested by Bonhommet and Babkine (1967).

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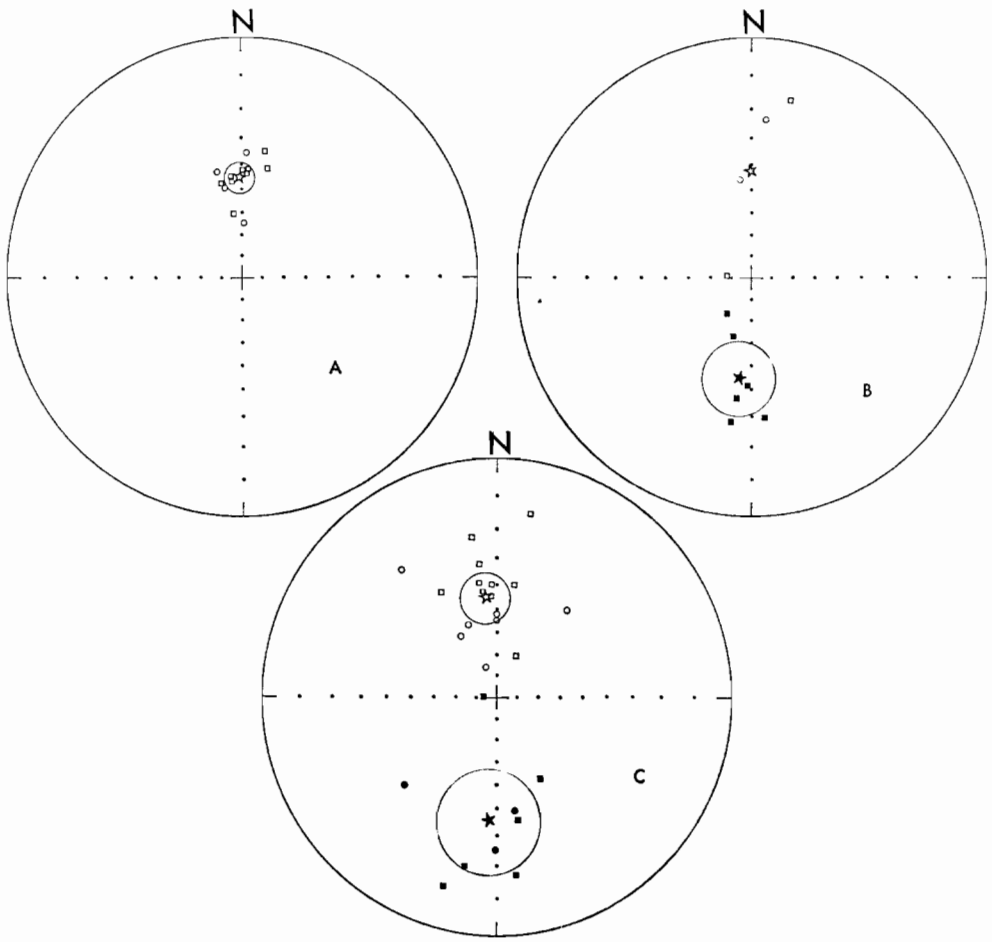


Figure 5. Stereographic projections of flow average directions of magnetization. (A) Late Lavas, Younger Volcanic Series. (B) Early Lavas, Younger Volcanic Series. (C) Older Volcanic Series. Open circles, upward directions. Closed circles, downward directions. Square symbols indicate dated lava flows. Stars represent mean directions and the larger circles are the corresponding 95 percent confidence limits.

Only two further results were obtained for the Matuyama reversed epoch. Both flows (MU 62, MU 63) have reversed polarity and their ages lie close to that of the Olduvai normal event. The question as to whether this is a double event awaits a detailed study of a stratigraphic sequence of flows extruded during this period of time.

Significant new results were obtained for the Gauss normal epoch. Two normally magnetized lavas, from each of which two samples were dated, yielded ages of high precision of 2.68 m.y. (MU 58, MU 59) and 2.74 m.y. (MU 49, MU 50). The results agree with sample W12 from New Mexico (Dalrymple and others, 1967) and help to set younger limits to the Kaena reversed event of McDougall and Chamalaun (1966). The Mammoth reversed event has been confirmed through two massive flows dated at 2.98 and 3.01 m.y. (MU 71, MU 70, MU 72). The basalt from the Sierra Navada (S21) from which Doell and others (1966) inferred the Mammoth event, was dated by them at 3.06 ± 0.1 m.y. Our results are not significantly different and together with data from Iceland (McDougall and Wensink, 1966) indicate a mean age of the Mammoth event of 3.03 m.y., with a duration possibly as short as 0.1 m.y.

Dates on two lavas with reversed polarity help to define the boundary between the Gauss normal and Gilbert reversed epochs. The two

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TABLE 5. STATISTICAL ANALYSIS OF PALEOMAGNETIC RESULTS AND MEAN DIRECTIONS

		N	R	K	α_{95}	δ	Mean Direction		Corresponding Pole	
							Declination	Inclination	Long.	Lat.
Late Lavas Younger Volcanic Series	Normal	13	12.7946	58.4349	5.4	10.1	359.3	−45.7	242.2	+83.1
Early Lavas Younger Volcanic Series	Normal	4	3.4963	5.9564	41.1	29.0	000.7	−42.8	228.5	+85.3
	Reversed	6	5.7330	18.7323	15.8	17.1	187.6	+46.6	020.3	−77.1
	All	10	9.2341	11.7511	14.7	22.5	004.9	−45.2	217.0	+78.1
Older Volcanic Series	Normal	17	75.9527	15.2785	9.4	20.2	355.2	−44.7	272.3	+82.5
	Reversed	9	8.0322	8.2667	19.0	26.8	183.8	+37.2	335.8	−86.3
	All	26	23.9040	11.9276	8.5	23.1	358.3	−42.2	257.2	+85.5

Notes: N Number of flows
R Resultant vector sum of N unit vectors
K Between flow precision parameter (Fisher, 1953)
α_{95} Angle of confidence at 95% probability level (Fisher, 1953)
δ Between flow angular standard deviation factor (Wilson, 1959)

dates of 3.34 and 3.46 m.y. (MU 74, MU 65, MU 69) together with previous results indicate that the age of the boundary is 3.35 ± 0.05 m.y. No information related to the Cochiti and Nunivak normal events (Dalrymple and others, 1967; Cox and Dalrymple, 1967b) in the Gilbert reversed epoch was obtained.

Results for flows dated between 4 and 8 m.y. are plotted in Figure 7. Because the uncertainty in the ages in this time range is relatively large the actual errors are shown in the figure. The results are not inconsistent with those obtained by Dalrymple and others (1967) for the same age range. This, however, is hardly unexpected in view of the paucity of the data. The Signal Mountain results form an impressive sequence of normally magnetized lavas with ages extending from 7.0 to 7.5 m.y. The period of normal polarity indicated by these results appears to be in remarkably good agreement with the period of normal polarity between 7.07 and 7.46 m.y. proposed by Heirztler and others (1968) on the basis of the sea-floor spreading hypothesis. However, this agreement may be fortuitous as the dates agree essentially to within experimental error and the time span covered by the sequence may actually be as small as 0.2 m.y.

As discussed earlier, some of the ages obtained are inconsistent with the field relationships. In the Grand Malabar Mountains this is probably due to excess radiogenic argon carried by the plagioclase, whereas in the Blanche Mountains it may be due to deuteric alteration resulting in loss of radiogenic argon. These results clearly highlight the necessity for making detailed studies on *stratigraphic* sequences of lava flows, rather than on isolated samples, so that the internal consistency can be checked. We feel that this is particularly important when attempting to extend the polarity time scale beyond 3.5 m.y. ago. The increasing amount of evidence that plagioclase may contain excess radiogenic argon (Damon and others, 1967), throws some doubt on the reliability of K-Ar results obtained on this mineral.

THE MASCARENE ISLANDS AND REGIONAL STRUCTURE

Mauritius, Réunion and Rodriguez together comprise the Mascarene Islands, which are aligned approximately in an east-northeasterly direction, transverse to and at the southern end of the Mascarene Plateau (Fig. 1). Rodriguez is built upon a guyot-like structure (McDougall and others, 1965) near the eastern extremity of the Rodriguez Ridge, which strikes essentially due west and intersects the Mascarene Plateau to the north of Mauritius. At the intersection the ridge is flanked by two deep passes separating it from Mauritius to the south and the Cargados Carajos Bank to the north (Fisher and others, 1967). Réunion is a separate volcanic edifice, 170 km west-southwest of Mauritius; between the two islands depths of more than 4000 m are recorded.

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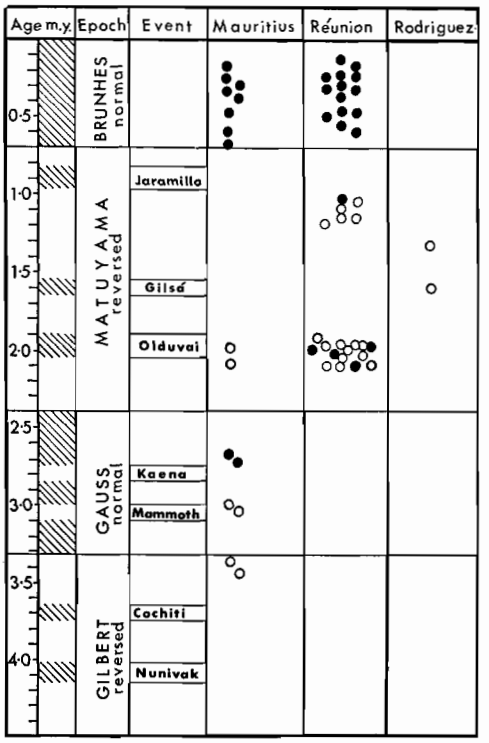


Figure 6. Diagram of indicated age and magnetic polarity of rocks from the Mascarene Islands between 0 and 4.5 m.y. Closed circles, normal polarity; open circles, reversed polarity.

Réunion consists of two volcanoes; Piton des Neiges, which occupies the northwestern part of the island, and the still active Piton de la Fournaise in the southeast.

In Figure 8 the evolution of the three islands is summarized. Only two dates of 1.3 and 1.5 m.y. are available from Rodriguez (McDougall and others, 1965; Upton and others, 1967), indicating that the nepheline bearing alkali olivine basalts which built the island were erupted during the period of quiescence between the outpouring of the Early and Late Lavas of the Younger Volcanic Series of Mauritius, and during a period of little activity on Réunion. The earliest activity recognized on Réunion comprises the Cirque Agglomerate of Piton des Neiges (Upton and Wadsworth, 1965), but no dating was possible on these rocks. Eruption of the Oceanite Series basalts followed between 2 m.y. and 0.45 m.y. ago (Chamalaun and McDougall, 1966), and these lavas constructed the main shield of the volcano. The lavas are transitional in character between tholeiites and alkali olivine basalts (Upton and Wadsworth, 1966). A veneer of alkali andesites and feldsparphyric

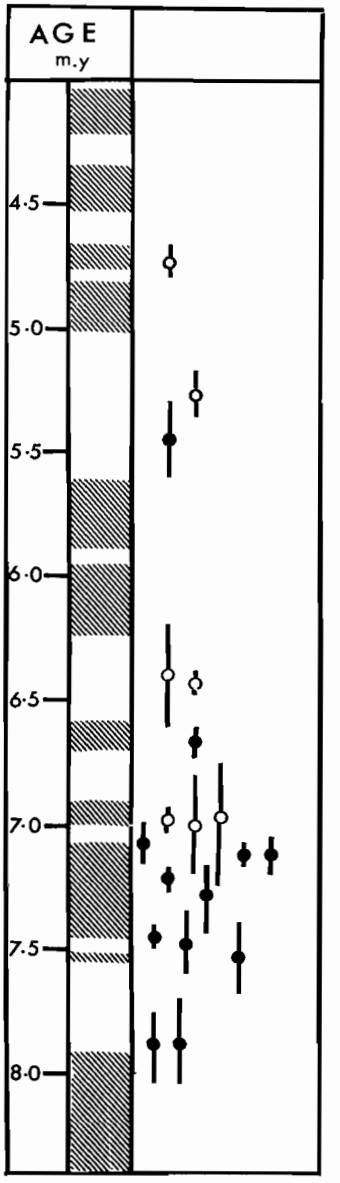


Figure 7. Diagram of indicated age and magnetic polarity of Mauritius rocks between 4.5 and 8.0 m.y. Closed circles, normal polarity; open circles, reversed polarity. Error in the age measurements at the 68 percent confidence level are indicated.

TABLE 6. MEAN AGES AND PALEOMAGNETIC DIRECTIONS FOR MAURITIUS LAVA FLOWS

Dated Sample	Age (m.y.)	Polarity	Mean Direction		No. Samples	Resultant Vector	Angular Standard Deviation (Wilson, 1959)	Corresponding Pole	
			Declination	Inclination				Long.	Lat.
Late Lavas of the Younger Volcanic Series									
MU 109	0.173 ± 0.003	N	353.0	−60.9	3	2.9820	6.1	251.1	+67.5
MU 27	0.26 ± 0.02	N	354.0	−44.5	2	1.9829	7.4	279.3	+81.9
MU 129	0.33 ± 0.01	N	001.6	−42.7	2	1.9989	1.8	219.3	+85.2
MU 24	0.34 ± 0.01	N	004.5	−42.0	5	4.9722	6.0	191.6	+84.2
MU 101	0.39 ± 0.05	N	354.5	−46.2	3	2.9570	9.7	270.7	+81.1
MU 118	0.47 ± 0.08	N	348.1	−47.9	2	1.9948	4.0	286.2	+76.1
MU 56, MU 57	0.61 ± 0.01	N	003.9	−43.0	3	2.8917	15.4	200.7	+84.0
MU 127	0.70 ± 0.01	N	011.1	−34.5	2	1.9789	8.3	142.3	+79.4
Early Lavas of the Younger Volcanic Series									
MU 61	1.96 ± 0.02	R	215.6	+70.4	2	1.9973	2.9	028.2	−46.3
MU 62, MU 63	2.08 ± 0.01	R	196.8	+62.4	3	2.9930	3.8	030.4	−62.6
MU 58, MU 59	2.67 ± 0.03	N	013.8	−16.2	1	—	—	107.4	+72.0
MU 49, MU 50	2.73 ± 0.01	(N)	277.1	−78.4	5	4.9149	10.5	261.3	+21.4
MU 70, MU 72	2.98 ± 0.04	R	184.2	+44.1	2	1.9970	3.1	023.5	−83.2
MU 71	3.01 ± 0.01	R	178.1	+38.1	1	—	—	112.1	−87.9
MU 74	3.33 ± 0.03	R	188.7	+28.5	2	1.9885	6.1	297.4	−80.2
MU 65, MU 69	3.44 ± 0.05	R	182.3	+34.7	5	4.9413	8.7	299.5	−87.4

TABLE 6. (*Continued*)

Dated Sample	Age (m.y.)	Polarity	Mean Direction		No. Samples	Resultant Vector	Angular Standard Deviation (Wilson, 1959)	Corresponding Pole	
			Declination	Inclination				Long.	Lat.
Older Volcanic Series: Port Louis Massif									
MU 7, MU 8	7.12 ± 0.04	N	357.8	−44.3	2	1.9951	3.9	256.2	+83.9
MU 130	7.13 ± 0.07	N	353.4	−42.1	2	1.9506	12.7	292.2	+82.7
MU 10	7.29 ± 0.15	N	011.6	−14.6	1	—	—	100.2	+72.9
MU 4	7.45 ± 0.08	N	332.1	−37.4	2	1.9804	8.0	321.0	+63.9
MU 1	7.22 ± 0.06	N	351.4	−39.0	2	1.9871	6.5	313.1	+81.8
MU 18	7.88 ± 0.13	N	358.2	−39.0	2	1.9796	8.1	279.3	+87.5
MU 15	7.49 ± 0.13	N	000.0	−51.6	1	—	—	237.5	+78.0
Older Volcanic Series									
MU 160	4.73 ± 0.06	R	150.5	+48.4	2	1.9838	7.2	122.5	−61.8
MU 169	5.26 ± 0.09	R	191.3	+19.9	2	1.9998	0.5	286.6	−75.1
MU 155	5.44 ± 0.14	N	351.5	−22.8	1	—	—	012.1	+78.2
MU 139, MU 140	6.40 ± 0.22	R	169.1	+36.0	2	1.9947	4.1	147.2	−79.8
MU 151	6.43 ± 0.02	(R)	272.8	+85.3	2	1.9492	12.9	047.7	−19.5
MU 41, MU 42	6.67 ± 0.08	N	040.3	−39.2	2	1.9664	10.5	158.3	+52.4
MU 141, MU 142	6.97 ± 0.26	R	170.3	+39.6	2	1.9931	4.7	131.7	−80.7
MU 43	6.99 ± 0.02	R	173.7	+17.9	2	1.9981	2.4	208.2	−77.3
MU 45	7.02 ± 0.02	R	196.4	+12.8	2	1.9661	10.5	288.8	−68.9
MU 171, MU 172	7.08 ± 0.07	N	026.0	−68.3	2	1.9967	3.2	210.7	+52.7
MU 133	7.53 ± 0.14	N	353.0	−31.4	2	1.9995	1.1	352.9	+82.5
MU 138	7.86 ± 0.18	N	010.5	−39.2	1	—	—	160.2	+82.0

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basalts, termed the Differentiated Series by Upton and Wadsworth (1965), covers much of the Oceanite Series of Piton des Neiges, and was erupted between about 0.35 and 0.10 m.y. ago (Chamalaun and McDougall, 1966). Piton de la Fournaise commenced eruption at least 0.3 m.y. ago, and produced lavas similar to those of the Oceanite Series. The volcanism in Réunion overlaps substantially in time with the Younger Volcanic Series of Mauritius, but occurred well after the main dome building phase of Mauritius was completed about 6.8 m.y. ago. Réunion clearly shows a progression with time of volcanic activity toward the southeast oblique to the trend of the Mascarene Islands and the Mascarene Plateau.

Although the Mascarene Islands are regarded as geographically related, Fisher and others (1967) concluded from the bathymetry that the three islands are independent topographic entities. This view is further supported by our results. From Figure 8 and the preceding discussion it is obvious that volcanism on the three islands has occurred more or less simul-

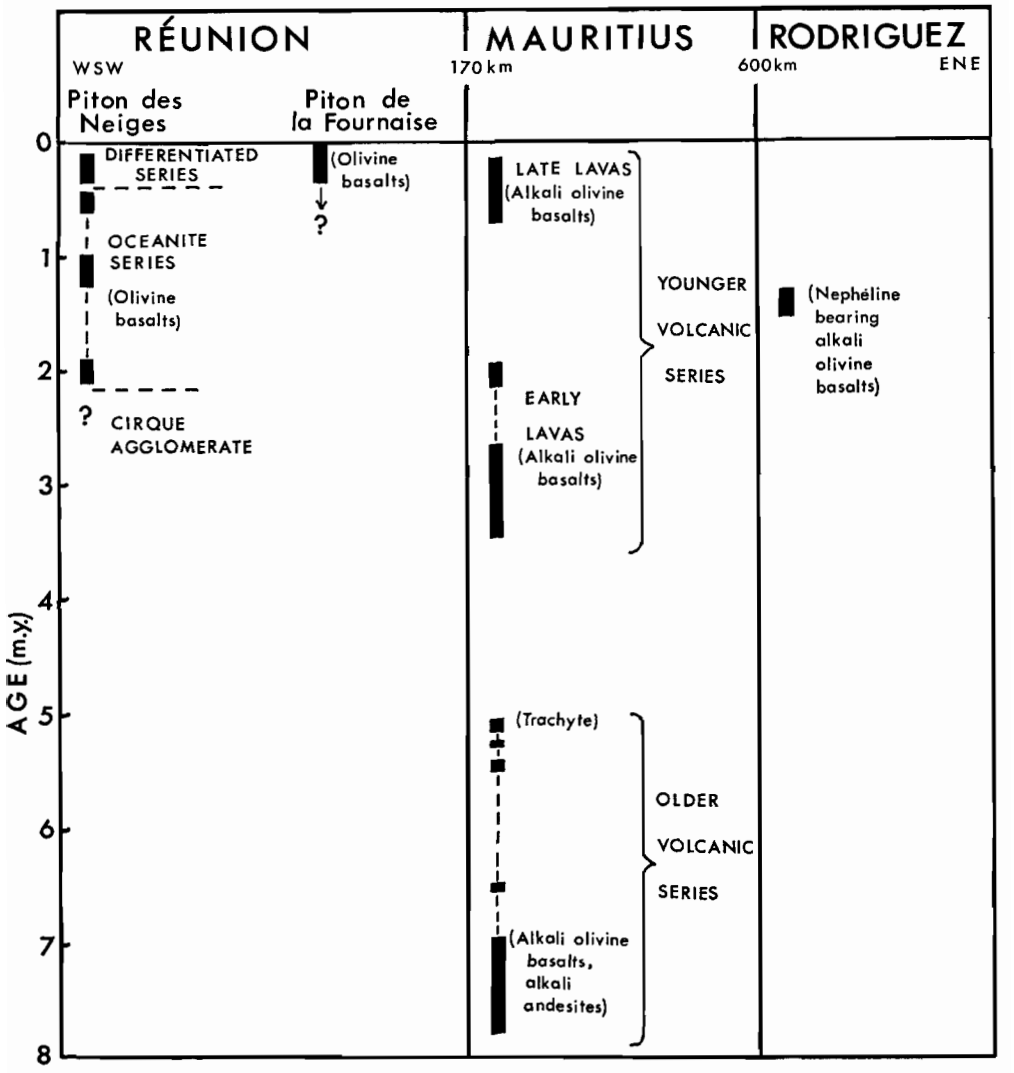


Figure 8. Summary of geochronology of the Mascarene Islands.

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taneously during the last 2 m.y., and that the lavas erupted are of different petrological types. We interpret this to indicate that each island acted as an independent center of eruption, and that for this period of time independent sources for the magmas must be postulated in the upper mantle, essentially directly beneath each volcano.

The dates obtained on lavas exposed in the volcanoes only provide minimum ages for the commencement of activity. However, because of the structural unity between the exposed parts of the volcanoes and their foundations, shown particularly well for Réunion and Mauritius (Fisher and others, 1967), it is suggested that each island as a whole is not much older than the dated subaerial activity.

Réunion and Mauritius are more than 1000 km west of the Mid-Indian Ocean Ridge. Hence, the hypothesis of Wilson (1963) that oceanic islands increase progressively in age with increasing distance from the mid-ocean ridges apparently does not apply to the Mascarene Islands. Le Pichon and Heirtzler (1968) interpret linear magnetic anomalies adjacent to the Mid-Indian Ocean Ridge in this region as indicating that sea-floor spreading has occurred from the southeast branch of the ridge for at least the last 10 m.y. and possibly for more than 30 m.y. The volcanism in the Mascarene Islands appears to be far too young to be related in any obvious way to this sea-floor spreading, although it is stressed that the ages found do not conflict with this concept.

The location of Mauritius and Réunion immediately to the south of the main Mascarene Plateau suggests that these volcanic islands may be more closely related to the development of this major structure. There is a trend of decreasing age of the main dome building lavas from northeast to southwest for Mauritius and Réunion, although the transverse trend in Réunion itself has already been pointed out. As noted earlier the Mascarene Plateau rises steeply from oceanic depths on either side. Geophysical evidence (Shor and Pollard, 1963; Davies and Francis, 1964; Matthews and Davies, 1966; Bunce and others, 1966) shows that the northern part of the plateau in the region of the Seychelles Archipelago has a thick continental type crust, and Precambrian granites have been recognized on Mahé (Miller and Mudie, 1961; Baker and Miller, 1963). However, from the Saya de Mahla Bank southward the Mascarene Plateau is structurally different; Bunce and others (1966) suggested that it is a volcanic ridge. Possibly the volcanism that has built Mauritius and Réunion is a continuation of the development of the Mascarene Plateau with migration of the activity generally southward with time. The Mauritius Trench may also be related to this development and the localization of the volcanism.

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APPENDIX

Detailed information on the localities from which samples were dated is given below and plotted in Figure 2. Grid references were read from the 1:100,000 map of Mauritius (Series Y682, edition 1, 1957, D., Survey War Office and Air Ministry, England). For those samples which are fresh and holocrystalline from thin section examination, no comment is made concerning their petrography.

Older Volcanic Series

MU 1 : EC 520693, Signal Mt. Track, Port Louis area.
MU 4 : EC 520691, Signal Mt. Track.
MU 7,8 : EC 507697, Signal Mt. Track.
MU 10 : EC 511694, Signal Mt. Track.
MU 14 : EC 563676, head of Vallée des Prêtres, below Pieter Both. Minor alteration to chloritic material in groundmass.
MU 15 : EC 563683, near MU 14 and petrographically similar.
MU 18 : EC 520664, south side of Anse Courtois at small dam. Incipient alteration of plagioclase, and some chlorite in groundmass.
MU 41, 42 : EC 440453, dike 2 m wide intruding strongly altered, amygdaloidal lavas in Black River. Incipient alteration of pyroxene and minor chlorite.
MU 43 : Dike 10 m wide, only 20 m from MU 41. Some replacement by chlorite.
MU 45 : EC 423473, mouth of Black River Gorge. Minor development of chlorite.
MU 80 : EC 394425, Chamarel trachyte plug, from top of eastern peak.
MU 81 : EC 393427, 200 m north of MU 80 in road cutting.
MU 116 : EC 610568, trachyte from northeast face of Piton du Milieu.
MU 117 : EC 607567, trachyte on south side of Piton du Milieu.
MU 130 : EC 513688, Quoin Bluff.
MU 133 : EC 459591, in waterfall between Corps de Garde and Mt. St. Pierre. Abundant olivine phenocrysts occur in slightly altered basaltic groundmass, with some areas of chlorite. Other lavas exposed here are strongly altered and zeolitized.
MU 138 : EC 479658, small quarry on north face of Petit Malabar. Minor chlorite in groundmass.
MU 139, 140 : EC 456638, at viaduct at western end of Grand Malabar Mountains. Except for alteration of olivine, the rock is fresh.
MU 141, 142 : Basalt immediately overlying MU 139. Large crystals of olivine, plagioclase, and augite occur in a holocrystalline, intergranular groundmass in which some chlorite occurs. The larger crystals sometimes occur in rounded aggregates and may well be xenolithic; this is supported by anomalously old date found for plagioclase (Table 2).
MU 151 : EC 912615, northwest face of Fayence Mountains, near Mare Jacquot. Groundmass shows incipient alteration.
MU 155 : EC 691593, west end of Blanche Mountain range. Strongly fluidal rock composed of poorly crystallized feldspar laths and alkalic pyroxene; small areas of brown to green alteration occur and also some calcite.
MU 160 : From thick horizontal lava on north side of Blanche Mts. near Hermitage. Small areas of green to brown fibrous alteration products occur in the groundmass.
MU 161 : EC 760581, from Deep River, near eastern end of Blanche Mountains. Microphenocrysts of olivine, partly altered, occur in slightly altered basaltic groundmass.

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MU 169 : EC 728575, eastern flank of Mt. Barbe, Blanche Mountains. Chloritic material in groundmass is common.

MU 171, 172 : EC 755481, from columnarly jointed flow on east flank of Lion Mt.

Early Lavas, Younger Volcanic Series

MU 49, 50 : EC 413477, lava at least 20 m thick, near junction of Petite Gorges and Black Rivers. Rare amygdules filled with chloritic material.

MU 58, 59 : EC 455344, Riviere Jacotet, under bridge at Beau Champ. Simpson (1951) mapped the basalts here as Late Lavas, but on basis of age they clearly belong to Early Lavas.

MU 61 : EC 378351, on road on west side of Baie du Cap.

MU 62, 63 : 100 m north of MU 61.

MU 65, 69 : EC 383352, from distinctive hackly jointed basalt at road level, east side of Baie du Cap. Groundmass is poorly crystallized, and may contain nepheline.

MU 70, 72 : EC 381346, eastern headland of Baie du Cap at road level. A few percent poorly crystallized material occurs in groundmass.

MU 71 : Same locality as MU 70, but probably one flow higher in sequence.

MU 74 : EC 405377, on Choisy to Chamarel road, altitude about 210 m.

MU 85 : EC 442420, Plaine Champagne, overlooking head of Black River Gorges.

Late Lavas, Younger Volcanic Series

MU 24 : EC 501701, outskirts of Port Louis at underpass on highway to Curepipe. Marginal alteration of olivine.

MU 27 : EC 511623, at River Cascade on Port Louis-Curepipe highway. Minor intersertal, poorly crystallized, feldspathic material.

MU 56, 57 : EC 471333, 100 m upstream from coast road, on Riviere des Galets. Fresh rock, petrographically more typical of the Early Lavas than the doleritic basalts of the Late Lavas; however, included with Late Lavas on basis of young age.

MU 101 : EC 797581, beneath bridge and near mouth of Grand River South East. Olivine partly altered to iddingsite.

MU 109 : EC 675884, north coast at St. Francois. About 5 percent intersertal, poorly crystallized material.

MU 118 : EC 451695, west coast at Pointe aux Sables.

MU 127 : EC 533318, south coast at Souillac. Minor intersertal, isotropic glass. Petrographically like Early Lavas, but included with Late Lavas because of young age.

MU 129 : EC 533307, south coast at Gris Gris, 2 km southeast of Souillac. Fresh rock with minor intersertal, isotropic glass. Petrographically similar to Early Lavas.